A Statistics Prospective in Digital Twins

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Digital twins

- **What are Digital Twins?**
  - The concept varies depending on individual perspectives and application domains.
  - Let's allow time to define what a good definition is – most likely, the group that succeeds in applying it most "effectively" will shape the prevailing understanding.
- **What Types of Mathematical/Statistical Models Are Important for Building or Applying Digital Twins?** Models that:
  - **Explore dynamic behavior** – capable of modeling time-dependent systems and evolving processes.
  - **Adapt in real-time** – make corrections based on incoming data.
  - **Learn from past observations** – incorporate learning mechanisms such as machine learning or Bayesian updates.
  - **Quantify discrepancies** – detect and measure deviations between the physical and digital models or high fidelity models and low fidelity models.

Using GP for emulators

- They are simple and easy to understand, and most importantly, they provide a well-behaved predictive posterior distribution that can be used as a surrogate.
- They are flexible and can accommodate a wide range of applications.
- Sometimes are misunderstood as restrictive.
- The mean and covariance functions can be modeled in complex ways, allowing for the handling of non-isotropy, non-stationarity, discontinuities, and more.
- Transformations can be efficiently applied to address non-Gaussian posterior errors.
- However, each attempt to address these challenges introduces computational complexity.

Multifidelity Computer Model Set-up

Consider S computer models $\{\mathscr{C}_t\}_{t=1}^{S}$

- aiming at simulating the same real system $\mathscr{S}_t$
- ordered by ascending fidelity level t.
- are expensive to run many times

Let

- $y_{tj}(x) : \mathcal{X} \rightarrow \mathbb{R}$ denote the output function of the computer model at coordinate j and code level t $\mathscr{C}_t$ where d_x-dimensional input $x \in \mathcal{X}$. For $j = 1, \ldots, n_s$

Consider:

- training data-set at level t as $(y_t, \mathfrak{X}_t)$,
- experimental design $\{\mathfrak{X}_t\}$ such that $x_{i,t} \in \mathfrak{X}_t$ for $i = 1, ..., n_t$ at t.

AR co-kriging (ARCK) statistical model: A review

- The auto-regressive co-kriging model (originally proposed by Kennedy O'Hagan (2000)):

$$y_t(x) = \xi_{t-1}(x) y_{t-1}(x) + \delta_t(x) \qquad \text{for } x \in \mathcal{X},\ t = 2, ..., S$$

where

  - $\{\delta_t(\cdot)\}$ and $\{\xi_t(\cdot)\}$, account for 'missing' or 'misrepresented' physical properties in the lower fidelity computer model $\mathscr{C}_{t-1}$ with respect to the higher one $\mathscr{C}_t$.
  - $\delta_t(\cdot)$ is the additive discrepancy function (representing the local adjustment from $\mathscr{C}_{t-1}$ to $\mathscr{C}_t$),
  - $\xi_t(\cdot)$ is the scale discrepancy (representing the scale change from $\mathscr{C}_{t-1}$ to $\mathscr{C}_t$ for $t = 1, ..., S$).

ADCIRC Storm Surge Application

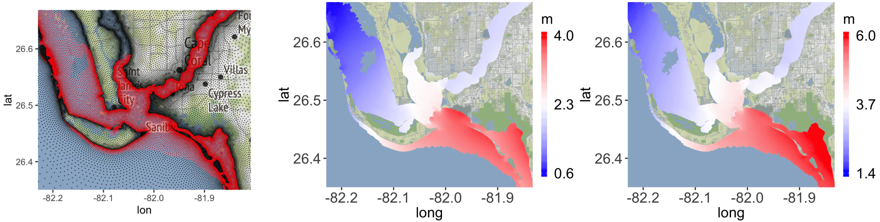

Figure: The mesh of the simulators with two different simulation runs for the storm surge ADCIRC simulator in Cape Coral (Fort Myers) Florida.

- ADCIRC is a computer model (used by FEMA) that simulates storm surge.
  - hydrodynamic circulation numerical model that solves the shallow-water equations for water levels and horizontal currents using the finite-element method.
  - Computational time for **peak surge** elevations over $9,284$ spatial location: 5 hours per run

⊖ Challenge in modeling & efficiency

- Treat large output dimensionality
  - Treat large output dimensionality;
  - Treating spatial coordinates as additional input requires large scale computations; Eg, inversion of matrices with sizes $9,284 \times (226+60) = 2,636,656$.
- Make it applicable to Non-hierarchically nested designs

$$\mathfrak{X}_{t+1} \not\subseteq \mathfrak{X}_t \text{ for some } t$$

–

- Efficient emulator

$$y_S(\cdot)|y \sim \mathrm{GP}(\mu^{\otimes S}(\cdot), C^{\otimes S}(\cdot,\cdot))$$

requires a matrix inversion of size $\sum_{t=1}^{S} n_t \times \sum_{t=1}^{S} n_t$.

Challenge & Model Specifications

- Treat large output dimensionality;
- Treating spatial coordinates as additional input requires large scale computations; Eg, inversion of matrices with sizes $9,284 \times (226+60) = 2,636,656$.
- Let N the number of spatial locations. For each coordinate $j = 1 : N$, we assume::
  - different regression parameters $\boldsymbol{\beta}_j := \{\boldsymbol{\beta}_{1,j}, \ldots, \boldsymbol{\beta}_{s,j}\}$
  - different variance parameters $\boldsymbol{\sigma}_j^2 := \{\sigma_{1,j}^2, \ldots, \sigma_{s,j}^2\}$
  - different scale discrepancy parameters $\boldsymbol{\xi}_j := \{\xi_{1,j}, \ldots, \xi_{t-1,j}\}$, where each $\xi_{1,j}$ is further assumed to be a unknown constant in the application.
  - $r(\cdot, \cdot|\phi_t)$ is a correlation function with correlation parameters $\phi_t := (\phi_{t,1}, \ldots, \phi_{t,d})^\top$ at fidelity level t.
  - The correlation parameters ϕ_t are assumed to be the same across different spatial coordinates to simplify computations and also add dependency within a fidelity level.

The statistical model

Coordinate $j = 1, ..., N$ and fidelity level $t = 2, \ldots, s$,

$$y_{t,j}(\mathbf{x}) = \gamma_{t-1,j}(\mathbf{x}) y_{t-1,j}(\mathbf{x}) + \delta_{t,j}(\mathbf{x})$$

Assign Gaussian processes priors on the unknown functions

$$\begin{aligned} y_{1,j}(\cdot) \mid \boldsymbol{\beta}_{1,j}, \sigma^2_{1,j}, \phi_1 &\sim GP(\mathbf{h}_1^\top(\cdot)\boldsymbol{\beta}_{1,j},\ \sigma^2_{1,j} r(\cdot,\cdot|\phi_1)), \\ \delta_{t,j}(\cdot) &\sim GP(\mathbf{h}_t^\top(\cdot)\boldsymbol{\beta}_{t,j},\ \sigma^2_{t,j} r(\cdot,\cdot|\phi_t)), \end{aligned}$$

Augment $\{y_{k,t}, \mathfrak{X}_{k,t}\}$ with missing data-set $\{\mathring{y}_{k,t}, \mathring{\mathfrak{X}}_{k,t}\}$ such that

$$L(\tilde{\mathbf{y}} \mid \boldsymbol{\beta}, \gamma, \boldsymbol{\sigma}^2, \phi) = \prod_{j=1}^{N} L(\tilde{\mathbf{y}}_j \mid \boldsymbol{\beta}_j, \gamma_j, \boldsymbol{\sigma}^2_j, \phi).$$

$$L(\tilde{\mathbf{y}}_j \mid \boldsymbol{\beta}_j, \gamma_j, \boldsymbol{\sigma}^2_j, \phi) \propto \pi(\tilde{\mathbf{y}}_{1,j} \mid \boldsymbol{\beta}_{1,j}, \sigma^2_{1,j}, \phi_1) \prod_{t=1}^{s} \pi(\tilde{\mathbf{y}}_{1,t} \mid \boldsymbol{\beta}_{1,t}, \sigma^2_{1,t}, \phi_t)$$

Set conjugate priors for $\{\boldsymbol{\beta}_{t,j}\}$ and $\{\gamma_t\}$, and proper priors for $\{\phi_t\}$

Model Specifications

Let y_{tj} be a vector of output values at coordinate j and code level t. For each coordinate j, we assume:

- different regression parameters $\boldsymbol{\beta}_j := \{\boldsymbol{\beta}_{1,j}, \ldots, \boldsymbol{\beta}_{s,j}\}$
- different variance parameters $\boldsymbol{\sigma}_j^2 := \{\sigma_{1,j}^2, \ldots, \sigma_{s,j}^2\}$
- different scale discrepancy parameters $\boldsymbol{\xi}_j := \{\xi_{1,j}, \ldots, \xi_{t-1,j}\}$, where each $\xi_{1,j}$ is further assumed to be a unknown constant in the application.
- $r(\cdot, \cdot|\boldsymbol{\phi}_t)$ is a correlation function with correlation parameters $\boldsymbol{\phi}_t := (\phi_{t,1}, \ldots, \phi_{t,d})^\top$ at fidelity level t.
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Prediction

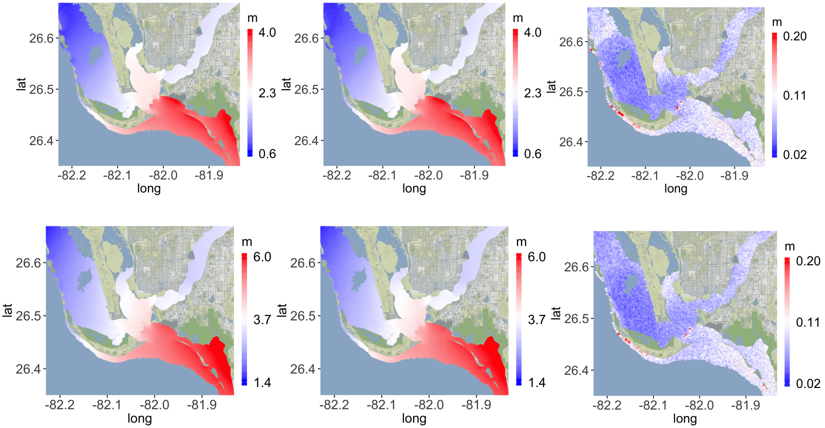


Figure: High-fidelity runs and predicted peak surge elevations with predictive standard errors at two input settings. High-fidelity runs, corresponding predicted PSE and the associated predictive standard errors.

Gaussian process based Transport Maps

- The transport map is an invertible and deterministic map between probability measures, typically bridging a target distribution and a reference distribution.
- $$T : \mathbb{R}^n \to \mathbb{R}^n \quad \text{such that} \quad x = T(z), \quad z \sim \eta$$
- Transport Map T is typically parametrized into a lower trianguler structure((Rosenblatt 1952):
- $$\mathcal{T}(y) = \begin{bmatrix} \mathcal{T}_1(y_1) \\ \mathcal{T}_2(y_1, y_2) \\ \vdots \\ \mathcal{T}_N(y_1, y_2, \ldots, y_N) \end{bmatrix}$$

Gaussian process based Transport Maps

- Define the transport map T in the following form(Matthias Katzfuss & Florian Schäfer (02 May 2023)):
- $$\mathcal{T}_i\left(y_{1:i}\right)=\left(y_i-f_i\left(y_{1:i-1}\right)\right)/d_i, \quad i=1,\ldots,N$$
- The distribution can be written as:
- $$p(y)=p_z(\mathcal{T}(y))|\det\nabla\mathcal{T}|=\prod_{i=1}^{N}\left(\mathcal{N}\left(\mathcal{T}_i\left(y_{1:i}\right)\mid 0,1\right)\left|\frac{\partial\mathcal{T}_i\left(y_{1:i}\right)}{\partial y_i}\right|\right)$$
- $$p(y)\propto\prod_{i=1}^{N}\mathcal{N}\left(y_i\mid f_i\left(y_{1:i-1}\right),d_i^2\right)$$

Gaussian process based Transport Maps

- Assigning NIG prior distribution to mean and variance:
- $$f_i \mid d_i \overset{\text{ind.}}{\sim} \mathcal{GP}\left(0,\ d_i^2 K_i\right), \quad i = 1, \ldots, N,$$ $$d_i^2 \overset{\text{ind.}}{\sim} \mathcal{IG}\left(\alpha_i, \beta_i\right), \quad \text{with } \alpha_i > 1,\ \beta_i > 0, \quad i = 1, \ldots, N.$$
- Where, $$K_i(\cdot,\cdot) = C_i(\cdot,\cdot)/\mathbb{E}(d_i^2), \quad \mathbb{E}(d_i^2) = \beta_i/(\alpha_i - 1)$$
- $$C_i\left(y_{1:i-1}, y'_{1:i-1}\right) = y_{1:i-1}^{\top} Q_i y'_{1:i-1} + \sigma_i^2 \rho_i\left(y_{1:i-1}, y'_{1:i-1}\right), \quad i = 1, \ldots, N$$

Multi-Fidelity Statistical Framework

Assume:

- Low-fidelity samples: $y_l = \{y_{i,l} \in \mathbb{R}^n\}_{i=1}^{n_L}$
- High-fidelity samples: $y_h = \{y_{i,h} \in \mathbb{R}^n\}_{i=1}^{n_H}$
- One complete sample: $y_m = (y_l, y_h)^T$
- Nested design: each HiFi sample has a matching LoFi input

Goal: Learn a joint transport map to model $p(y_l, y_h)$, or use conditional maps $T : x \mapsto y$

$$\mathcal{T}(y_m) = \begin{bmatrix} \mathcal{T}_l(y_l) \\ \mathcal{T}_h(y_l, y_h) \end{bmatrix}$$

Multi-Fidelity Transport Maps

Assume:

- High-fidelity data only depends on previous HiFi neighbors and also the LoFi data at the same location.
- Which is $T_{h,i}(y_{h,i}) = \frac{y_{h,i} - f_{h,i}(y_{h,1:i-1}, y_{l,i})}{d_{h,i}}$
- The Covariance function of the mean prior is :
$$C_{h,i}\left(y_{h,i,neigh}, y'_{h,i,neigh}\right) = y_{h,1:i-1}^{\top} Q_{h,i} y'_{h,1:i-1} + y_{l,i}^{\top} Q_{l,i} y'_{l,i} + \sigma_{h,i}^2 \rho_{h,i}\left(y_{h,1:i-1}, y'_{h,1:i-1}\right) + \tau_{h,i}^2 \rho_{h,i}\left(y_{l,i}, y'_{l,i}\right), \quad i = 1, \ldots, N,$$
- Where $y_{h,i,neigh} = (y_{h,1:i-1}, y_{l,i})$

Multi-Fidelity Transport Maps

Posterior map and integrated likelihood

- $$p(Y \mid f, d) = \prod_{i=1}^{N} \mathcal{N}_{n_l}\left(y_{l,i} \mid f_{l,i}, d_{l,i}^2 I_{n_l}\right) \times \mathcal{N}_{n_h}\left(y_{h,i} \mid f_{h,i}, d_{h,i}^2 I_{n_h}\right)$$

Priors:

- $$\begin{aligned} f_{l,i} \mid d_{l,i} &\overset{\text{ind.}}{\sim} \mathcal{GP}\left(0,\ d_{l,i}^2 K_{l,i}\right), \quad i = 1, \ldots, N, \\ d_{l,i}^2 &\overset{\text{ind.}}{\sim} \mathcal{IG}\left(\alpha_{l,i}, \beta_{l,i}\right), \quad \text{with } \alpha_{l,i} > 1,\ \beta_{l,i} > 0, \quad i = 1, \ldots, N, \\ f_{h,i} \mid d_{h,i} &\overset{\text{ind.}}{\sim} \mathcal{GP}\left(0,\ d_{h,i}^2 K_{h,i}\right), \quad i = 1, \ldots, N, \\ d_{h,i}^2 &\overset{\text{ind.}}{\sim} \mathcal{IG}\left(\alpha_{h,i}, \beta_{h,i}\right), \quad \text{with } \alpha_{h,i} > 1,\ \beta_{h,i} > 0, \quad i = 1, \ldots, N \end{aligned}$$

Marginal and Conditional Structure

The integrated likelihood can be derived as the product of the predictive distribution of each dimension given by all previous dimensions.

Integrated likelihood:

$$p(Y) \propto \prod_{i=1}^{N} t_{2\alpha_{l,i}}\left(0,\ \frac{\beta_{l,i}}{\alpha_{l,i}} G_{l,i}\right) \times t_{2\alpha_{h,i}}\left(0,\ \frac{\beta_{h,i}}{\alpha_{h,i}} G_{h,i}\right)$$

The two fidelity models are estimated *separately*, avoiding joint inversion.

Posterior Map Formulation

Two fidelities are modeled independently:

Low-Fidelity:

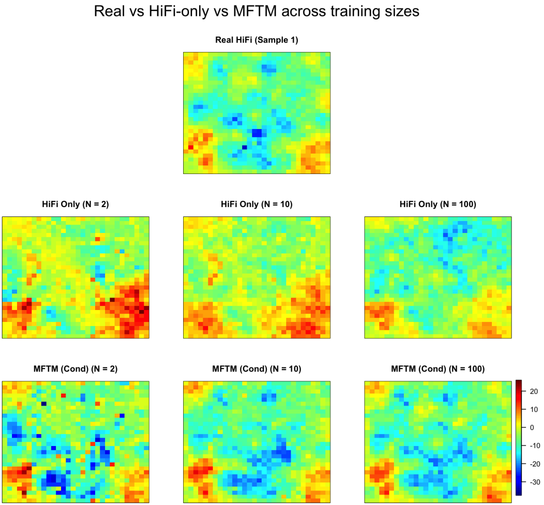
$$y^*_{l,i} = \hat{f}_{l,i}(y^*_{l,1:i-1}) + F^{-1}_{2\tilde{\alpha}_{l,i}}(\Phi(z^*_i))\, \hat{d}_{l,i} \left[v_{l,i}(y^*_{l,1:i-1}) + 1\right]^{1/2}, \quad i = 1, \ldots, N$$

High-Fidelity:

$$y^*_{h,i} = \hat{f}_{h,i}(y^*_{h,1:i-1},\, y^*_{l,i}) + F^{-1}_{2\tilde{\alpha}_{h,i}}(\Phi(z^*_i))\, \hat{d}_{h,i} \left[v_{h,i}(y^*_{h,1:i-1},\, y^*_{l,i}) + 1\right]^{1/2}$$

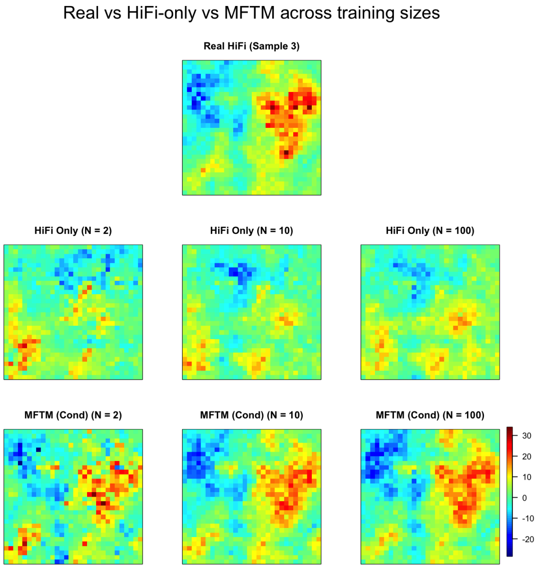
- $\hat{f}_{(\cdot)}$: posterior mean function
- $\hat{d}_{(\cdot)}$: posterior scale
- $v_{(\cdot)}$: posterior predictive variance adjustment
- F^{-1}_{ν}: inverse CDF of Student-t with ν d.o.f.

Simulation Results: Samples Comparison



Comparison of Real HiFi, HiFi-only, and MFTM (Cond) predictions for $N = 2, 10, 100$.

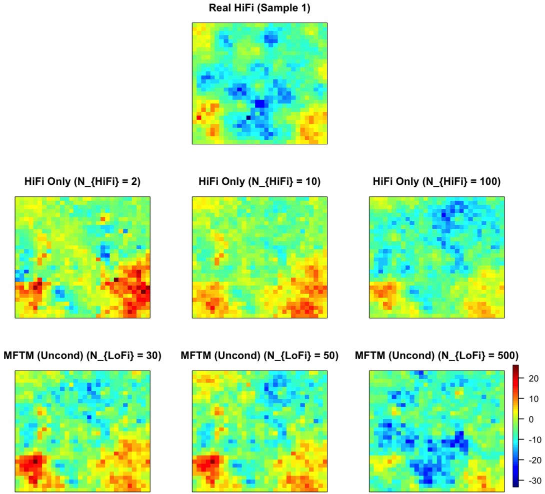
Simulation Results: Samples Comparison



Comparison of Real HiFi, HiFi-only, and MFTM (Cond) predictions for $N = 2, 10, 100$

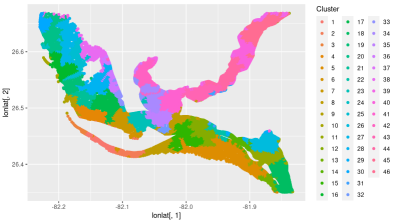
Simulation Results: Samples Comparison

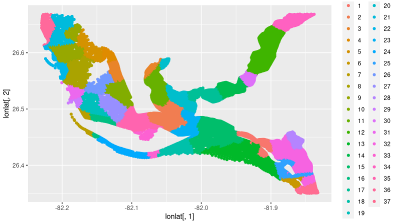
Real vs HiFi-only vs MFTM across training sizes



Comparison of Real HiFi, HiFi-only, and MFTM (Cond) predictions for $N = 2, 10, 100$.

Application Spanning Tree





	PP-Cokriging	Spatially Clustered cokriging (spatial points)	Spatially Clustered cokriging (nodes used)
RMSE	2.09	0.041	0.038
Runtime(h)	11	14	12

Thanks for your attention!